

3. Sign your application with the developer certificate that you got from enrolling in the iPhone Developer program. Deploy and test it on your device.
4. Submit your application to App Store for Apple's approval. App Store is a service for the iPhone and iPod Touch created by Apple that allows users to browse and download applications from the iTunes Store that was developed with the iPhone SDK and published through Apple. The apps are available for purchase or free of charge, depending on the application. App Store is available from both iTunes and iPhone.
5. If approved, you can see your application in the App Store. If you are selling a paid application, Apple will keep 30 per cent of sales revenue and give you 70 per cent. Figure 3 showcases the Indian App Store that you can access from iPhone.

That is the best way to go, if you have the money and have development on your mind. But that involves purchasing Mac hardware. As they say, if you spend money, you get money.

### iPhone development: The open way

Even with all these cool features, let's understand this: the factory-shipped iPhone is a closed device, like any other consumer computing device. If we talk about 'closedness', iPhone generally falls under the game console category along with Xbox 360, PlayStation 3, Sony PSP and Nintendo Wii. All these devices are very powerful computing devices. But the systems developers want to restrict the applications that are run on them. This gives them more control over the software and hence helps them to channelise the revenue generated. Most of the time, this is implemented with code signatures, where the signing key is only available to the hardware makers. Whatever applications run on these platforms have to be signed by the systems makers, and only then will they run on those systems. The reason is quite obvious.

Now, allow me to clarify: code signing in itself is not a bad thing. Code signing helps users to verify the source of the application and hence keep them safe from malicious code. In Linux, RPM repositories use GPG-based signatures to verify the integrity of RPM source. Windows uses device driver signatures to ensure the stability of a driver. The only difference is that even if something is not signed, it will run on Linux and Windows systems, whereas on consumer devices like the iPhone, it will simply not run.

iPhone lives in a 'jail', literally! In the UNIX world we called it the *chroot* jail. All iPhone interfacing applications, such as iTunes, run in a *chroot* environment where no application can peek into the operating system. Also, the applications that run on iPhone must be signed by Apple in order to run.

Before we proceed, let's try to understand *chroot* jails. *chroot*ing is a verb that evolved from the *chroot* (2) system call, which is used to change the root of the file system as seen by the calling process. When a process requests to *chroot* to a given directory, any future system calls issued by

the process will see that directory as the file system root. It becomes impossible to access files and binaries outside the tree rooted on the new root directory. This environment is known as a *chroot* jail.

A jail is a directory in your system, and the user cannot see/do anything outside that directory. In other words, the user is jailed in the directory. The *chroot* (2) system call is used to put the user inside this jail. If you want the user to be able to do one thing, that one thing should be possible inside the jail. For example, if you want the user to be able to run SCP, you need to have a copy of SCP installed in the jail, in addition to the means to execute it (a shell).

In order to run your own application on iPhone (without Apple's signature), you must free your iPhone from this 'jail', commonly referred to as jail-breaking your iPhone.

 **Warning:** I am not a lawyer. Before jail-breaking your iPhone, please check whether this renders your warranty as null and void. A Google search might help.

You will need a Windows or Mac system to jail break your device. It is a very easy process. Please visit iPhone dev team's blog at [blog.iphone-dev.org/post/74278878/close-the-stable-door](http://blog.iphone-dev.org/post/74278878/close-the-stable-door) to understand the process. Download a tool called QuickPwn, a very self-explanatory tool that will help you render your iPhone as jail broken within minutes.

### Setting up your iPhone for development

Jail-breaking your iPhone will give you a lot of UNIX utilities installed on the iPhone, including BSD user land applications. Jail-breaking also installs a Debian-based GUI package management tool called Cydia (refer to the sidebar title 'Cydia Package Manager' for more information on this utility). Open Cydia and search for software called



Figure 3: Indian App Store on iPhone

### Cydia package manager

Quoting Wikipedia: "Cydia is a Debian Apt-based package installer/manager for the iPhone and iPod Touch, created and maintained by Jay Freeman (saurik), which allows users to browse and download applications from a range of sources. Most applications are available to download for free, with some requiring purchase after downloading, and as of the 1.0.2790-44 release, applications are available to purchase from within the application." For more details visit [en.wikipedia.org/wiki/Cydia\\_\(iPhone\\_OS\)](http://en.wikipedia.org/wiki/Cydia_(iPhone_OS)).